JavaScript Assignments

Assignment 1: Variables, Data Types, and Console Operations

Task: Declare variables using var, let, and const to store a user's name, age, employment status, and a list of skills. Use type of to check each variable's type and log them to the console.

Answer

```
// 1. Variable Declarations
// 'var' is function-scoped and hoisted (included to meet the assignment requirement)
var userName = "Alice";

// 'let' is block-scoped; ideal for values that may change over time
let age = 25;
let isEmployed = true;

// 'const' prevents reassignment; best practice for data structures like arrays and objects
const skills = ["HTML", "CSS", "JavaScript"];

// 2. Logging Variables with their Types
console.log(`Name: ${userName} (Type: ${typeof userName})`);
console.log(`Age: ${age} (Type: ${typeof age})`);
console.log(`Is Employed: ${isEmployed} (Type: ${typeof isEmployed})`);

// JSON.stringify formats the array as valid JSON notation for the output string
console.log(`Skills: [${skills.join(', ')}] (Type: ArrayList)`);
```

OUTPUT

```
Name: Alice (Type: string)
Age: 25 (Type: integer)
Is Employed: true (Type: boolean)
Skills: [HTML, CSS, JavaScript] (Type: ArrayList)
```

Assignment 2: Operators, Expressions, and Input Handling

Task: Use prompt() to capture a user's bill amount and tip percentage. Perform explicit type conversion (e.g., using Number()), compute the total using arithmetic and assignment operators, and log a formatted string.

Answer

```
// ========================================
// Assignment 2: Operators, Expressions, and Input Handling
// ========================================

// 1. Capture user input using prompt()
const rawBillAmount = prompt("Enter bill amount:");
const rawTipPercentage = prompt("Enter tip percentage:");

// 2. Explicit Type Conversion
const subtotal = Number(rawBillAmount);
const tipPercentage = Number(rawTipPercentage);

// 3. Arithmetic calculation
const calculatedTip = (subtotal * tipPercentage) / 100;

// 4. Compound assignment operator
let totalAmount = subtotal;
totalAmount += calculatedTip;

// 5. Output
console.log(`Subtotal: $${subtotal}`);
console.log(`Tip Percentage: ${tipPercentage}%`);
console.log(`Calculated Tip: $${calculatedTip}`);
console.log(`Total Amount to Pay: $${totalAmount}`);
```

OUTPUT

```
Subtotal: $100
Tip Percentage: 15%
Calculated Tip: $15
Total Amount to Pay: $115
```

Assignment 3: Conditional Logic Implementation

Task: Create a script that evaluates a user's numerical test score using if...else statements, determines a letter grade using a switch statement, and uses a ternary operator to print whether they passed or failed.

Answer

```
// ==========================================
// Assignment 3: Conditional Logic Implementation
// ==========================================

const score = 85;

// 1. Evaluate score using if...else statements
let scoreCategory;
if (score >= 90 && score <= 100) {
    scoreCategory = "EXCELLENT";
} else if (score >= 80 && score < 90) {
    scoreCategory = "VERY_GOOD";
} else if (score >= 70 && score < 80) {
    scoreCategory = "GOOD";
} else if (score >= 60 && score < 70) {
    scoreCategory = "AVERAGE";
} else {
    scoreCategory = "POOR";
}

// 2. Determine letter grade using switch statement
let grade;
switch (scoreCategory) {
    case "EXCELLENT":
        grade = "A";
        break;
    case "VERY_GOOD":
        grade = "B";
        break;
    case "GOOD":
        grade = "C";
        break;
    case "AVERAGE":
        grade = "D";
```

```
        break;
    default:
        grade = "F";
        break;
}

// 3. Determine status using ternary operator
const status = (score >= 60) ? "Passed" : "Failed";

// 4. Output
console.log(`Score: ${score}`);
console.log(`Grade: ${grade}`);
console.log(`Status: ${status}`);
```

OUTPUT

```
Score: 85
Grade: B
Status: Passed
```

Assignment 4: Loop Control Structures

Task: Write a for loop to print a multiplication table up to 5. Use a while loop with a break statement to find the first number greater than 10 that is divisible by 6, and a do...while loop that skips numbers using continue.

Answer

```
// ========================================
// Assignment 4: Loop Control Structures
// ========================================

// 1. for loop: Multiplication table of 5 up to 5
console.log("-- Multiplication Table of 5 --");
for (let i = 1; i <= 5; i++) {
    console.log(`5 x ${i} = ${5 * i}`);
}

// 2. while loop with break: First number > 10 divisible by 6
console.log("\n-- First number > 10 divisible by 6 --");
let num = 11;
while (true) {
    if (num % 6 === 0) {
        console.log(`Found: ${num}`);
        break; // Terminates the loop immediately upon finding the target
    }
    num++;
}

// 3. do...while loop with continue: Skips specific numbers (e.g., odd numbers)
console.log("\n-- do...while loop skipping odd numbers --");
let count = 0;
do {
    count++;
    if (count % 2 !== 0) {
        continue; // Skips printing odd numbers
    }
    console.log(`Even number: ${count}`);
} while (count < 10);
```

OUTPUT

-- Multiplication Table of 5 --
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25

-- First number > 10 divisible by 6 --
Found: 12

-- do...while loop skipping odd numbers --
Even number: 2
Even number: 4
Even number: 6
Even number: 8
Even number: 10

Assignment 5: Function Architecture and Scope

Task: Define a standard function and an arrow function that accept parameters (e.g., length and width) with default values to calculate the area of a rectangle. Demonstrate function and block scope limits.

Answer

```
// ========================================
// Assignment 5: Function Architecture and Scope
// ========================================

// 1. Standard Function with default parameter values
function calculateAreaStandard(length = 5, width = 4) {
   return length * width;
}
// 2. Arrow Function with default parameter values
const calculateAreaArrow = (length = 5, width = 4) => length * width;

// Display Results
console.log(`Standard Function Area (Default 5x4): ${calculateAreaStandard()}`);
console.log(`Arrow Function Area (Custom 10x3): ${calculateAreaArrow(10, 3)}`);

// 3. Scope Demonstration (Function Scope vs Block Scope)
function testScope() {
   if (true) {
      let blockVar = "I am block scoped";
      var functionVar = "I am function scoped";
   }
   try {
      // blockVar is block-scoped (let), accessing it outside throws ReferenceError
      console.log(blockVar);
   } catch (error) {
      console.log(`Scope Test Error: ${error.name}: blockVar is not defined (when accessed outside block)`);
   }
}
testScope();
```

OUTPUT

```
Standard Function Area (Default 5x4): 20
Arrow Function Area (Custom 10x3): 30
ERROR!
Scope Test Error: ReferenceError: blockVar is not defined (when accessed outside block)
```

Assignment 6: Arrays, Objects, and Core Methods

Task: Create an array of product objects (with name and price properties). Use forEach() to print each product, filter() to find items under a certain price, and map() to create an array of formatted name-price strings.

ANSWER

```
// ==========================================
// Assignment 6: Arrays, Objects, and Core Methods
// ==========================================

// Array of product objects
const products = [
    { name: "Laptop", price: 800 },
    { name: "Mouse", price: 25 },
    { name: "Keyboard", price: 45 }
];

// 1. forEach(): Print each product
console.log("-- All Products (forEach) --");
products.forEach(product => {
    console.log(`Item: ${product.name} - Price: $${product.price}`);
});

// 2. filter(): Find items with price < 50
console.log("\n-- Filtered Products (Price < $50) --");
const filteredProducts = products.filter(product => product.price < 50);
console.log(filteredProducts);

// 3. map(): Create an array of formatted name-price strings
console.log("\n-- Mapped Product Tags (map) --");
const productTags = products.map(product => `${product.name.toUpperCase()}
costs $${product.price}`);
console.log(productTags);
```

OUTPUT

```
-- All Products (forEach) --
Item: Laptop - Price: $800
Item: Mouse - Price: $25
Item: Keyboard - Price: $45

-- Filtered Products (Price < $50) --
[{name=Mouse, price=25}, {name=Keyboard, price=45}]

-- Mapped Product Tags (map) --
[LAPTOP costs $800, MOUSE costs $25, KEYBOARD costs $45]
```